Bioaccumulation and biomagnification

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In the same way that nutrients and other compounds can be cycled through an ecosystem, so can harmful toxins and pollutants. The terms *bioaccumulation* and *biomagnification* are often interchanged but each has a specific meaning. Both words refer to the effects of pollutants (or toxins) entering the food chain. The difference lies in their effects once they are present.

Bioaccumulation describes how substances enter the food chain at the lowest trophic level, in other words, from the grazer upward to the final trophic level, the top carnivore. Bioaccumulation describes the accumulation of an undesirable substance *within the tissues of an organism or a population of organisms.*

Biomagnification (sometimes called *bioconcentration*) describes the magnified effects, or increasing levels of concentration, of these compounds *as they travel along the food chain* (trophic levels). Consider the simplified example below used to illustrate how the amount of toxin can become biomagnified within an ecosystem:

> *A certain water bug is considered a pest in Lake Perfect. The water is treated with a toxic chemical compound that kills most of the bugs. One molecule of poison is sufficient to kill each bug, therefore each dead bug contains one molecule of the toxin.*
>
> *Ten 'mini-fish' come along and each fish eats 50 poisoned bugs. The tissues of each mini-fish then contain 50 molecules of toxin. A large 'killer fish' then eats 10 of the mini-fish (each of which contains 50 molecules of toxin). The tissues of the killer fish then contain 500 molecules of toxin.*
>
> *A predatory bird eats 5 of the killer fish. Its tissues will contain 2500 molecules of toxin. A land carnivore kills and eats 3 of the predatory birds (each of which has eaten 5 killer fish). The tissues of the land predator will contain 7500 molecules of toxin.*

Thus, the compound used to kill the water bugs was biomagnified from one molecule per individual, at the lowest trophic level, to 7500 per individual at the top level as it travelled up the food chain. In some instances, substances can bioaccumulate at one level only but the organism's metabolism is able to break it down. Because of this, it may not become biomagnified throughout the whole food chain.

1 Show in figures how, in the example above, one molecule of toxin magnified to 7500.

Give meanings, in biological terms, for these words:

2 uptake

3 ingest

4 storage

5 eliminate

6 equilibrium

7 Describe the difference between *bioaccumulation* and *biomagnification.*

8 Suggest other ways in which scientists might look for evidence of bioaccumulation besides testing animal tissue.

9 A certain branch of science is called *ecotoxicology.* Suggest what ecotoxicologists do.

Identify the following terms mentioned on this sheet.

10 A hierarchical layer of a food web.

11 Flesh-eating mammal.

12 A substance that provides energy and vital substances to an organism.

13 Combination of organisms and the area in which they live.

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